Lecture 15: Alternating Series and Conditional Convergence

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Learning Outcome

At the end of this lecture, you should be able to:

- ☐ Understand what is an alternating series.
- ☐ Apply the Alternating Series Test to determine if an alternating series converges.
- ☐ Apply the Alternating Series Estimation Theorem to estimate the sum of a convergent alternating series.
- ☐ Understand what is a conditionally convergent series.
- ☐ Understand how rearranging an infinite series affect its sum.

Alternating series

A series in which the terms are alternatively positive and negative is called an $\sum a_n$ alternating series.

Example:

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = \frac{1}{1} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \cdots$$

$u_n = \frac{1}{n}$; +ve, +ve, −ve, +ve, −ve, +ve ...

← convergent alternating harmonic series.

$$\sum_{n=1}^{\infty} \frac{(-1)^n 4}{2^n} = -2 + 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots$$

$u_n = \frac{4}{2^n}$

← geometric series with $r = -\frac{1}{2}$ $(|r| < 1)$
So the series converges.

$$\sum_{n=1}^{\infty} (-1)^{n+1} n = 1 - 2 + 3 - 4 + 5 - 6 + 7 - 8 + \cdots$$

$u_n = n$

$1 + (-2+3) + (-4+5) + (-6+7) + \cdots$
$+1 \quad +1 \quad +1$

(divergent)

Alternating Series Test

The Alternating Series Test

The series

$$\sum_{n=1}^{\infty}(-1)^{n+1}u_n = u_1 - u_2 + u_3 - u_4 + \cdots$$

converges if the following conditions are satisfied:

1. The u_n's are all positive. $u_n > 0$
2. The u_n's are eventually nonincreasing. There must exist an integer N such that $u_n \geq u_{n+1}$ for all $n \geq N$
3. $u_n \to 0$. $\lim_{n\to\infty} u_n = 0$

Proof: Assume $N=1$ ($u_n \geq u_{n+1}$ for all $n \geq 1$)

For even n, $n = 2m$ for some integer m. ← definition of an even number.

The partial sum $s_n = s_{2m} = (u_1 - u_2) + (u_3 - u_4) + \ldots + (u_{2m-1} - u_{2m})$

≥ 0 $\quad \geq 0$ $\quad \geq 0$

So $\{S_{2m}\}$ should be a nondecreasing sequence.

Now notice that at the same time, we can also write

$$S_{2m} = u_1 - u_2 + u_3 - u_4 + u_5 - \dots - u_{2m-2} + u_{2m-1} - u_{2m}$$

$$= u_1 - \underbrace{(u_2 - u_3)}_{\geq 0} - \underbrace{(u_4 - u_5)}_{\geq 0} - \dots - \underbrace{(u_{2m-2} - u_{2m-1})}_{\geq 0} - \underbrace{(u_{2m})}_{>0}$$

So $S_{2m} < u_1$ for all m.

So $\{S_{2m}\}$ is bounded from above by u_1

So $\lim_{n \to \infty} S_n = L < \infty$ (converges)

Alternating Series Test

For odd n, $n = 2m+1$ for some integer m.

$$s_n = s_{2m+1} = \underbrace{u_1 - u_2 + u_3 - u_4 + \cdots + u_{2m-1} - u_{2m}}_{s_{2m}} + u_{2m+1}$$

$$= s_{2m} + u_{2m+1}$$

$$\lim_{n\to\infty} s_{2m+1} = \lim_{n\to\infty} s_{2m} + \lim_{n\to\infty} u_{2m+1} = L.$$

$\lim_{n\to\infty} s_{2m} \to L$, $\lim_{n\to\infty} u_{2m+1} \to 0$

So, the series converges.

Alternating Series Test

Example: $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n}$ (Alternating harmonic series)

$u_n = \frac{1}{n}$ (>0, >0)



① $u_n > 0$ for all $n \geq 1$

② $u_n \geq u_{n+1}$ for all $n \geq 1$

③ $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{n} = 0$

By Alternating Series Test, $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n}$ converges.

The Alternating Series Test

Example: $\sum_{n=1}^{\infty}(-1)^{n+1}\frac{10n}{n^2+16}$ u_n.

$u_n = \frac{10n}{n^2+16}$

① $u_n > 0$ for all $n \geq 1$

② Let $f(x) = \frac{10x}{x^2+16}$

$$f'(x) = \frac{10(x^2+16) - 2x(10x)}{(x^2+16)^2}$$

$$= \frac{10(x^2 - 2x^2 + 16)}{(x+16)^2}$$

$$= \frac{10(16-x^2)}{(x^2+16)^2} \leq 0 \text{ if } x \geq 4$$

($10 > 0$, $(x^2+16)^2 > 0$)

≤ 0 for $x \geq 4$

✓ u_n is non-increasing for $n \geq 4$.

✓ ③ $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{10n}{n^2+16}$

$$= \lim_{n\to\infty} \frac{\frac{10}{n} \to 0}{1 + \frac{16}{n} \to 0} = 0$$ ✓

So, by Alternating Series Test,

the series converges.

Error in Approximating Alternating Series

Suppose that a_k is a sequence satisfying

$$a_0 \geq a_1 \geq \cdots \geq a_n \geq a_{n+1} \geq \cdots \geq 0,$$

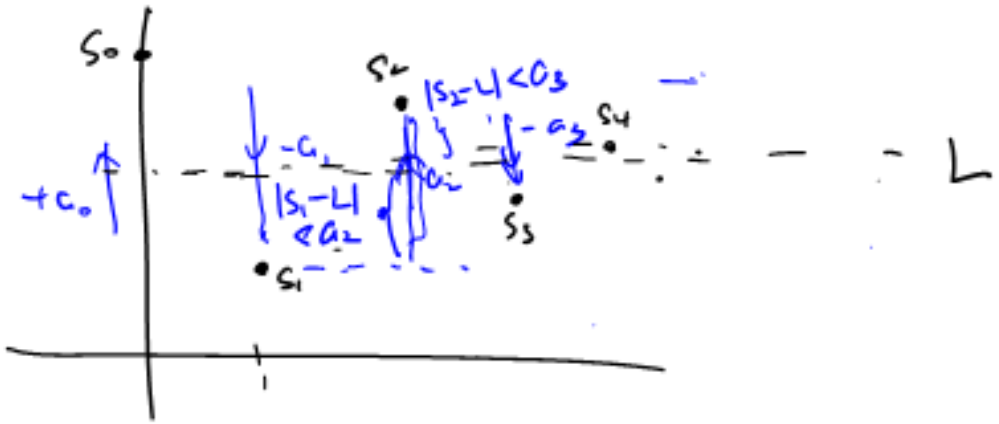
non-increasing; positive

Satisfying all three conditions in AST.

with $\lim_{n\to\infty} a_n = 0$.

So $\sum (-1)^n a_n = L$ (converges).

We let $s_n = \sum_{k=0}^{n} (-1)^k a_k = a_0 - a_1 + a_2 - a_3 + a_4 - a_5 + \cdots$



$|s_n - L| < a_{n+1}$

L is always between s_n and s_{n+1}.

Alternating Series Estimation Theorem

The Alternating Series Estimation Theorem

If the alternating series $\sum_{n=1}^{\infty}(-1)^{n+1}u_n$ satisfies the three conditions of the Alternating Series Test, then for $n \geq N$, ← the N on the condition.

$$s_n = u_1 - u_2 + \cdots + (-1)^{n+1}u_n$$

approximates the sum L of the series with an error whose absolute value is less than u_{n+1}, the absolute value of the first unused term. Furthermore, the sum L lies between any two successive partial sum s_n and s_{n+1}, and the remainder, $L - s_n$, has the same sign as the first unused term.

$s_n < L < s_{n+1}$ or $s_{n+1} < L < s_n$

$$|L - s_n| < u_{n+1}$$

$$-u_{n+1} < L - s_n < u_{n+1}$$

$$L > s_n - u_{n+1} \qquad L < s_n + u_{n+1}$$

If n is even, $S_n = u_1 - u_2 + u_3 - u_4 + \ldots + u_{n-1} - u_n$

$$S_{n+1} = u_1 - u_2 + u_3 - u_4 + \ldots + u_{n-1} - u_n + u_{n+1}$$

$$= S_n + u_{n+1}$$

$$S_n < L < S_{n+1}$$

If n is odd, $S_n = u_1 - u_2 + u_3 - u_4 + \ldots - u_{n-1} + u_n$

$$S_{n+1} = S_n - u_{n+1}$$

$$S_{n+1} < L < S_n$$

Alternating Series Estimation Theorem

Example: $\sum_{n=0}^{\infty}(-1)^n \frac{1}{2^n}$ $= 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \cdots$ $a = 1$

u_n

geometric series with $r = \frac{-1}{1}$

(convergent)

$$\sum_{n=0}^{\infty}(-1)^n \frac{1}{2^n} = \frac{1}{1-\left(\frac{-1}{1}\right)} = \frac{2}{3} \approx 0.66667.$$

Ex: $S_8 = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} + \frac{1}{64} - \frac{1}{128} \approx 0.6640625$

$S_9 = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} + \frac{1}{64} - \frac{1}{128} + \frac{1}{256} \approx 0.667968$

It is clear to see that $S_8 < 0.66667 < S_9$

and $0.66667 - S_8 < \frac{1}{256}$ $\leftarrow u_9$

Conditional Convergence

A series that is convergence but not absolutely convergent is called conditional convergent.

$\sum a_n$ converges but $\sum |a_n|$ diverges

Eg: Alternating harmonic series. converges conditionally.

$$\sum (-1)^{n+1} \frac{1}{n} \quad \text{converges} \quad (\text{AST})$$

but $\sum \left| (-1)^{n+1} \frac{1}{n} \right| = \sum \frac{1}{n}$ diverges (harmonic series)

Conditional Convergence

Example: For a positive constant p, consider the alternating p-series

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^p}.$$

Let $a_n = \frac{(-1)^{n+1}}{n^p}$

$\sum |a_n| = \sum \frac{1}{n^p}$

For $p>1$, $\sum |a_n|$ is a convergent p-series.

$\Rightarrow \sum a_n$ converges.

For $0 < p \leq 1$

$\sum |a_n|$ is a divergent p-series.

What about $\sum a_n = \sum \frac{(-1)^{n+1}}{n^p}$?

Let $u_n = \frac{1}{n^p}$

✓ ① $u_n > 0$ for all $n \geq 1$

✓ ② $u_n \geq u_{n+1}$ for all $n \geq 1$

✓ ③ $\lim_{n\to\infty} u_n = \lim_{n\to\infty} \frac{1}{n^p} = \left(\lim_{n\to\infty} \frac{1}{n}\right)^p = 0$ ($\frac{1}{n} \to 0$)

So by Alternating Series Test, $\sum a_n$ converges.

For $p > 0$, the alternating p-series $\begin{cases} \text{converges absolutely if } p > 1 \\ \text{converges conditionally if } 0 < p \leq 1 \end{cases}$

Rearranging Series

The Rearrangement Theorem for Absolutely Convergent Series

If $\sum_{n=1}^{\infty} a_n$ converges absolutely, and $b_1, b_2, \ldots, b_n, \ldots$ is any arrangement of the sequence $\{a_n\}$, then $\sum_{n=1}^{\infty} b_n$ converges absolutely and another series with same the same term but different arrangement as $\sum a_n$.

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} a_n.$$

So we can always rearrange the terms of an absolutely convergent infinite series. But can we do the same for a conditional convergent series?

Rearranging Series

Consider the alternating harmonic series (conditional convergent series)

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \dots = L.$$

If we rearrange the term, for example in the following way.

$$\left(1 - \frac{1}{2}\right) - \frac{1}{4} + \left(\frac{1}{3} - \frac{1}{6}\right) - \frac{1}{8} + \left(\frac{1}{5} - \frac{1}{10}\right) - \frac{1}{12} + \left(\frac{1}{7} - \frac{1}{14}\right) - \frac{1}{16} + \dots$$

$$= \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \frac{1}{14} - \frac{1}{16} + \dots$$

$$= \frac{1}{2}\left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \dots\right)$$

$= L$.

$$= \frac{1}{2}L$$ But clearly that $L \neq \frac{1}{2}L$.

!!! So NEVER simply rearrange the terms of a conditionally convergent series, especially in finding the limit!

Summary

$a_n \not\to 0 \Rightarrow$ the series diverges.

1. **The *n*th-Term Test for Divergence:** Unless $a_n \to 0$, the series diverges.
2. **Geometric series:** $\sum ar^n$ converges if $|r| < 1$; otherwise it diverges.
3. ***p*-series:** $\sum 1/n^p$ converges if $p > 1$; otherwise it diverges.
4. **Series with nonnegative terms:** Try the Integral Test or try comparing to a known series with the Direct Comparison Test or the Limit Comparison Test. Try the Ratio or Root Test.
5. **Series with some negative terms:** Does $\sum |a_n|$ converge by the Ratio or Root Test, or by another of the tests listed above? Remember, absolute convergence implies convergence.
6. **Alternating series:** $\sum a_n$ converges if the series satisfies the conditions of the Alternating Series Test.